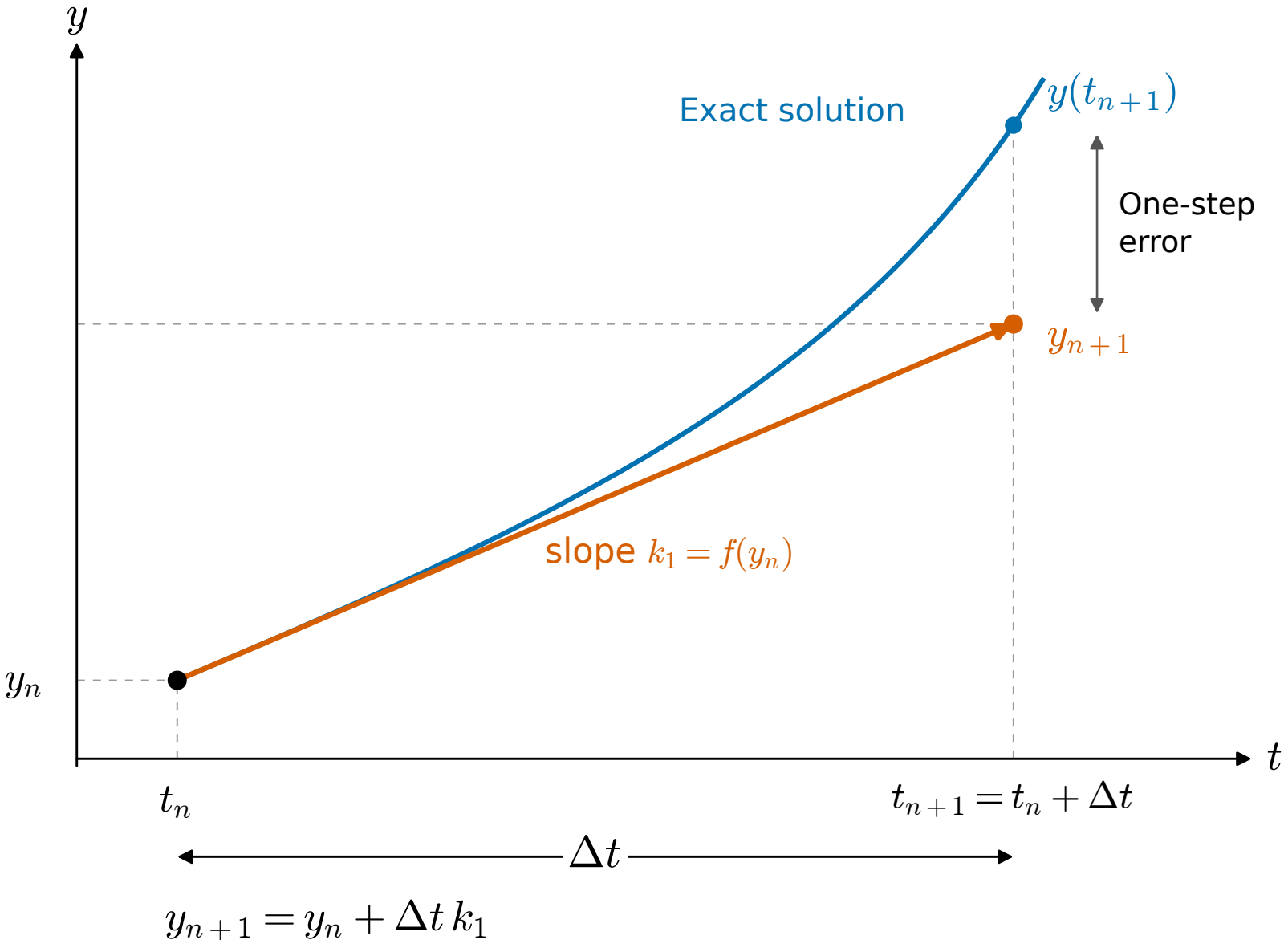
Euler: use the slope at the start

$y' = 1 + y^2, \quad t_n = 0, \quad y_n = 0, \quad \Delta t = 1$



$$y_{n+1} = y_n + \Delta t\, k_1$$